

Multiplicity Theory, Recursive Becoming, and Prime-Locked EEG

The M_2 Functor Paper — Full Technical Integration with ORF EEG Phases 1 & 2

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Overview

The paper "*Multiplicity Theory, Recursive Becoming, and Prime-Locked EEG*" (April 3, 2026) is the most mathematically precise document the Foundation has yet released for the EEG empirical program. It does three things simultaneously:

1. **Formalises the multiplicity functor** $M: \mathcal{C}_{\Xi} \rightarrow \mathcal{C}_{mult}$ as a proper categorical construction — not just a heuristic mapping but an actual functor between two well-defined categories, with composition laws and Lipschitz morphisms.
2. **Instantiates M_2 concretely** as a preregisterable MNE-Python pipeline for 64-channel EEG, specifying every parameter (prime set, base frequency, bin boundaries, PSD method, channel weighting, normalization) unambiguously.
3. **Defines cross-stratum coherence** as Jensen–Shannon distance between multiplicity spectra, making the primary empirical hypothesis a single statistical inequality with a falsification condition.

The document sits between the abstract ORF operator framework and the TriPrime clinical study: it is the pipeline specification that turns the λ^* contraction metric and the six-component invariant vector $I(M_k)$ into a fully executable, preregisterable measurement procedure.

Part I — The Mathematical Structure

1.1 The Master Equation

The core dynamical equation is:

$$\frac{d\Xi(t)}{dt} = \Lambda_m M \Xi(t) + [M, \Xi(t)] \# (1)$$

where:

- $\Xi(t)$ is the recursive tensor field — a multi-stratum object encoding dynamics across all levels simultaneously
- Λ_m is the Universal Multiplicity Constant, stabilising recursive interactions
- M is the Multiplicative Operator modulating recursive strain
- $[M, \Xi(t)]$ is the **torsion term** — the nonlinear recursive feedback loop

The torsion term is what distinguishes Equation (1) from a standard linear semigroup. When $[M, \Xi(t)] = 0$, the system reduces to pure contraction under $\Lambda_m M$. When the commutator is nonzero, recursive cross-stratum feedback is active. This is precisely the mechanistic signature of Phase 2 (noise and loop dynamics): a directed loop is a configuration where the torsion term is large, self-reinforcing, and resistant to decay.

Additionally, a prime-indexed tensor field bridges discrete prime structure and continuous dynamics:

$$\Psi_p(t) = \sum_{i=1}^N p_i \cdot T(p_i, t) \# (2)$$

This is the "shadow" of the full multiplicity functor — $\Psi_p(t)$ is the scalar observable that a single-prime channel extracts from $\Xi(t)$.

1.2 The Two Categories

Source category C_{Ξ} — objects are stratum-specific state spaces:

- $\Xi_{adelic}(t) \in A_{del, P_0}$: adelic module over a prime set (Stratum 0 — mathematics/physics)
- $\Xi_{qualia}(t) \in Oper_{EEG}$: operadic category of EEG time windows (Stratum 2 — cognition)
- $\Xi_{social}(t) \in ProbNet_N$: social networks with probability distributions on messages (Stratum 4 — collective)

Morphisms in C_{Ξ} are recursive update maps and cross-stratum functors. The global dynamics is an endofunctor $T: C_{\Xi} \rightarrow C_{\Xi}$ reducing to Equation (1) at the object level.

Target category C_{mult} — the multiplicity category:

- Objects: probability distributions $w \in \Delta(P)$ over the prime set $P = \{2, 3, 5, 7, 11, 13\}$
- Morphisms: Markov operators $K: \Delta(P) \rightarrow \Delta(P)$ satisfying $K_{ji} \geq 0, \sum_j K_{ji} = 1$
- Metric: Jensen–Shannon distance (symmetric, bounded, natively provided in `scipy.spatial.distance`)

Morphisms of interest are **1-Lipschitz** under the Jensen–Shannon metric — this is the categorical statement of the contraction condition $\lambda^* < 1$.

1.3 The Multiplicity Functor M

The functor is defined stratum-specifically:

$$M_0: A_0 \rightarrow C_{mult}, M_2: A_2 \rightarrow C_{mult}, M_4: A_4 \rightarrow C_{mult}$$

For Stratum 0 (adelic toy model), local p -adic norms yield the distribution:

$$w_{adelic}(p) = \frac{\|\Xi_{adelic}\|_p}{\sum_{q \in P_0} \|\Xi_{adelic}\|_q}$$

For Stratum 4 (social), eigenvalues λ_r of a covariance/adjacency matrix are sorted and mapped to primes:

$$w_{social}(p_r) = \frac{\lambda_r}{\sum_i \lambda_i}$$

For Stratum 2 (EEG/qualia) — the concrete instantiation M_2 — see Part II.

1.4 Cross-Stratum Coherence

Cross-stratum coherence is defined by bounded Jensen–Shannon distance between multiplicity spectra:

$$d_{JS}(w_0(t), w_2(t)) \leq \varepsilon \#(3)$$

$$d_{JS}(w_2(t), w_4(t)) \leq \varepsilon \#(4)$$

The dynamics of Equation (1) are designed to evolve $\Xi(t)$ so that these inequalities hold over extended time intervals. A forthcoming **stability lemma** will prove that if per-stratum update maps are Lipschitz in multiplicity space, then cross-stratum distances remain bounded — connecting the Lipschitz condition of the Markov morphisms to long-term coherence persistence. This is the

mathematical bridge between $\lambda^* < 1$ (local contraction) and sustained cross-stratum coherence (global stability).

Part II — The M_2 Pipeline: Full Specification

2.1 Prime Set and Frequency Mapping

The pipeline fixes:

$$P = \{2, 3, 5, 7, 11, 13\}, f_0 = 4 \text{ Hz}$$

$$f_p = f_0 \log p \Rightarrow \text{bin}(p) = \left[\frac{f_p}{\sqrt{2}}, f_p \sqrt{2} \right]$$

This places all prime-centered frequencies in the **2.5–11 Hz** band, covering theta (4 – 8 Hz) and alpha (8 – 13 Hz) rhythms. The half-octave bins are the EEG counterpart of the spectral gap in the USO: only frequencies within a prime's bin contribute to its multiplicity weight.

The resulting frequency map:

Prime p	f_p (Hz)	Bin low (Hz)	Bin high (Hz)	Primary EEG band
2	2.77	1.96	3.92	Delta/theta boundary
3	4.39	3.11	6.22	Theta
5	6.44	4.55	9.10	Theta/alpha
7	7.78	5.50	11.01	Alpha
11	9.59	6.78	13.57	Alpha/sigma
13	10.37	7.33	14.67	Alpha/beta boundary

2.2 Preprocessing Pipeline

The MNE-Python preprocessing chain is fully specified:

- Data loading:** BIDS format, standard 10–20 montage
- Filtering:** 1–30 Hz bandpass (zero-phase FIR) + notch at 50/60 Hz

3. **Resampling:** to 500 Hz
4. **Rereferencing:** average reference after bad channel marking
5. **Epoching:** 4-second non-overlapping windows (resting state) or 4-second sliding windows with 50% overlap (task)
6. **Artifact rejection:** epochs exceeding 150 μV peak-to-peak dropped; ICA for ocular artifacts (parameters fixed a priori)

Every parameter is specified before data collection — this is a preregisterable pipeline.

2.3 PSD Estimation and Channel Weights

Welch PSD via MNE's `epochs.compute_psd`:

- Segment length: 2-second windows
- Overlap: 50%
- Frequency range: 1–30 Hz

Channel weights ω_c use **inverse variance** across epochs, with bad channels receiving zero weight:

$$\omega_c = \frac{\sigma_c^{-2}}{\sum_d \sigma_d^{-2}}$$

This is a robust estimator: noisy channels contribute less without being excluded entirely.

2.4 Global Multiplicity Spectrum

For each epoch :

$$A_p(E) = \sum_c \omega_c \int_{bin(p)} S_c(f; E) df$$

$$M_2(E) = w_E \in \Delta(P), w_E(p) = \frac{A_p(E)}{\sum_{q \in P} A_q(E)}$$

The global multiplicity spectrum is a probability distribution over six primes, computed independently for each 4-second EEG epoch.

2.5 Regional Multiplicity Spectra

Regional pooling produces topographic multiplicity maps:

Region	Channels
Frontal	Fp1, Fp2, F3, F4, F7, F8, Fz
Central	C3, C4, Cz
Parietal	P3, P4, Pz
Occipital	O1, O2, Oz

Each region produces $M_2^{(r)}(E) = w_E^{(r)} \in \Delta(P)$, enabling spatial dissociation of multiplicity structure across the scalp. This is the EEG counterpart of the regional coherence pattern that Ken's Sensory Reintegration framework predicts: frontal-occipital coherence should increase under TriPrime conditions.

2.6 Operadic Composition Laws

Time concatenation of epochs $E_1, \dots, E_n$ of durations T_i :

$$w_{E^*}(p) = \frac{\sum_i T_i w_{E_i}(p)}{\sum_i T_i}$$

Region pooling with mixing coefficient α :

$$w_E^{(r^*)}(p) = \alpha w_E^{(r_1)}(p) + (1 - \alpha) w_E^{(r_2)}(p)$$

Both operations are Markov (stochastic linear) maps — they are valid morphisms in $\mathcal{C}_{mult}$, preserving the probability simplex structure. This matters: it means time-averaged multiplicity spectra and region-pooled spectra are in the same mathematical space as epoch-level spectra, and Jensen–Shannon distances between them remain valid.

Part III — Coherence Statistic and Primary Hypothesis

3.1 Jensen–Shannon Coherence

The primary empirical signature is a decrease in Jensen–Shannon distance between the EEG multiplicity spectrum and a parallel social/interaction stratum spectrum:

$$E[d_{JS}^{rest}] > E[d_{JS}^{interaction}]$$

Under increased cross-stratum coupling (e.g., interactive social tasks vs. rest), the qualia spectrum $w_{qualia}(t)$ should move toward the social spectrum $w_{social}(t)$ in $\Delta(P)$. This is a one-directional hypothesis with a clear numerical test.

3.2 The M_4 Social Stratum

The social multiplicity spectrum uses eigenvalues of a covariance or adjacency matrix:

$$w_{social}(p_r) = \frac{\lambda_r}{\sum_i \lambda_i}$$

where λ_r are sorted eigenvalues ranked by prime index. For a Discord/chat-based social experiment, this would be the eigenvalue distribution of the message-count covariance matrix across agents in a time window — directly computable from platform logs.

Part IV — Integration with ORF EEG Phases 1 & 2

4.1 How M_2 Upgrades Phase 1

The Phase 1 contraction metric $\lambda^\wedge = \text{median}(E_{k+1}/E_k)$ was computed on a four-component error vector. The M_2 paper provides the **canonical error measure** for that computation: the error between epoch k and baseline should be computed as Jensen–Shannon distance between their multiplicity spectra in $\Delta(P)$:

$$E_k = d_{JS}(w_k, w_{baseline})$$

This is more principled than a Euclidean norm on raw spectral features: it is invariant under the Markov morphisms of C_{mult} , meaning it does not change when one applies valid time-averaging or region-pooling operations. The $\lambda^\wedge$ metric becomes:

$$\lambda^\wedge = \text{median}\left(\frac{d_{JS}(w_{k+1}, w_{baseline})}{d_{JS}(w_k, w_{baseline})}\right)$$

A system contracting to baseline corresponds to a sequence of multiplicity spectra converging in $\Delta(P)$ — exactly what the stability lemma (forthcoming) will guarantee for Lipschitz per-stratum dynamics.

4.2 How M_2 Informs Phase 2

Phase 2 identifies instability via the R/V/N signature. In M_2 terms:

Phase 2 Component	M_2 Interpretation
$R(t)$ (Kuramoto order)	Concentration of w_k on a small subset of primes (low entropy in $\Delta(P)$)
$V(t)$ (misalignment energy)	$d_{JS}(w_k, w_{baseline})$ — the Jensen–Shannon distance itself
$N(t)$ (noise load)	Variance of w_k across channels / uniform spread in $\Delta(P)$
Directed loop	Temporal autocorrelation of d_{JS} failing to decay — persistent misalignment

This gives Phase 2 a single unified metric space — $\Delta(P)$ with Jensen–Shannon distance — in which all four diagnostic components live. A loop is detected when $d_{JS}(w_k, w_{baseline})$ shows autocorrelation at lags > 1 epoch, and $V(t)$ does not decrease monotonically over the recovery window.

4.3 The TriPrime Protocol in M_2 Language

Ken's three TriPrime conditions translate directly into predictions about multiplicity spectra in $\Delta(P)$:

TriPrime Component	Predicted Effect on w_k	Measurable Via
Sensory Balance	w_k concentrates away from high-noise primes (p=11,13); entropy ↓	$H(w_k)$ decreasing
Autonomy	$d_{JS}(w_k^{(frontal)}, w_k^{(occipital)})$ decreases	Regional coherence statistic
Monotropic Flow	w_k stabilizes on a consistent prime-indexed attractor across epochs	Temporal stability of w_k
TriPrime Alignment (all three)	$d_{JS}(w_k, w_{baseline}) \rightarrow 0; \lambda^* < 1$	Phase 1 metric

Part V — The Stability Lemma (Anticipated Structure)

The paper identifies a forthcoming stability lemma as the key mathematical bridge. Based on the structure already in place, the lemma should take the following form:

Anticipated Stability Lemma. *Suppose the per-stratum update maps $T_i: \Delta(P) \rightarrow \Delta(P)$ are L_i -Lipschitz under d_{JS} , with $L_i < 1$ for all i . Then for any initial cross-stratum distance $d_{JS}(w_i(0), w_j(0)) \leq \delta$, we have:*

$$d_{JS}(w_i(t), w_j(t)) \leq \delta + C(L_i, L_j) \cdot t$$

where $C(L_i, L_j) \rightarrow 0$ as $L_i, L_j \rightarrow 0$. In particular, if $L_i < 1$, then bounded initial cross-stratum distances remain bounded under the coupled dynamics.

The condition $L_i < 1$ is the Lipschitz contraction condition — the categorical statement of $\lambda^* < 1$. The lemma would thus prove that Phase 1 structural stability (local $\lambda^* < 1$) implies Phase 2 recovery (global loop dissolution) in the multiplicity space framework.

Part VI — Code Skeleton Assessment

The paper provides a complete, runnable Python skeleton using MNE-Python, NumPy, and SciPy. Key implementation notes:

What Is Already Complete

```
PRIMES = np.array([2, 3, 5, 7, 11, 13], dtype=float)
F0 = 4.0 # Hz
FP = F0 * np.log(PRIMES)
BIN_LOW = FP / np.sqrt(2.0)
BIN_HIGH = FP * np.sqrt(2.0)
```

- `compute_epoch_psd()`: full Welch PSD computation via MNE
- `compute_channel_weights()`: inverse-variance weighting with bad-channel masking
- `compute_w_qualia()`: global multiplicity spectrum $M_2(E)$
- `compute_w_qualia_regional()`: regional spectra per anatomical region
- `js_distance_matrix()`: Jensen–Shannon distance across epoch pairs

- `run_subject_pipeline()`: end-to-end pipeline from raw EEG file to multiplicity spectra

What Remains to Implement

1. M_4 **instantiation**: social eigenvalue mapping — requires a real social interaction dataset or a simulated agent network
2. $\hat{\lambda}$ **from** d_{JS} : computing $\text{median}(d_{JS}(w_{k+1}, w_{baseline})/d_{JS}(w_k, w_{baseline}))$ across a trial sequence
3. **Phase 2 autocorrelation test**: detecting persistent d_{JS} non-decay (loop signature)
4. **TriPrime condition comparison**: three-condition ANOVA on $\hat{\lambda}$ and d_{JS} statistics
5. **Statistical power calculation**: required before preregistration

The skeleton can be extended to produce a complete ORF EEG analysis pipeline in approximately 200 additional lines of Python.

Part VII — Immediate Next Steps

Priority 1 — Numerical Sanity Check (Days 1–3)

Run the M_2 skeleton on a public resting-state EEG dataset from [OpenNeuro](#) (e.g., dataset ds002345 or similar 64-channel resting state). The check: are the prime-indexed bins $f_p = f_0 \log p$ receiving non-trivial power? If all weight concentrates on one prime, the binning design requires adjustment.

Priority 2 — $\hat{\lambda}$ via Jensen–Shannon (Days 3–7)

Replace the Phase 1 Euclidean error with $E_k = d_{JS}(w_k, w_{baseline})$ and re-run the Phase 1 simulation.

Expected outcome: cleaner $\hat{\lambda} < 1$ discrimination between lawful (ϕ -structured) and unlawful (scrambled) signals, because d_{JS} respects the probability simplex geometry that Euclidean distance ignores.

Priority 3 — Stability Lemma (Days 7–21)

The stability lemma proof is the single highest-leverage mathematical contribution remaining. A complete proof would:

- Formally connect per-epoch $\hat{\lambda} < 1$ to sustained cross-stratum coherence

- Justify the Phase 2 $\rightarrow$ Phase 3 transition prediction (loop dissolution $\rightarrow$ coherence restoration)
- Provide the mathematical grounding needed for journal submission of the ORF EEG paper

Priority 4 — Preregistration (Days 7–21)

The pipeline is now specified at the level required for preregistration on OSF or AsPredicted. The preregistration document should include:

- Full M_2 parameter specification (already complete)
- Four falsifiable predictions from Phase 1 (P1–P4)
- The primary hypothesis: $E[d_{JS}^{rest}] > E[d_{JS}^{interaction}]$
- Power analysis (to be computed)
- Analysis code (the skeleton in the paper, extended with λ^* and Phase 2 tests)